



Cell Injury and Cell Death

Part I

AMS-HIS
Pathology Team





Session's Objectives

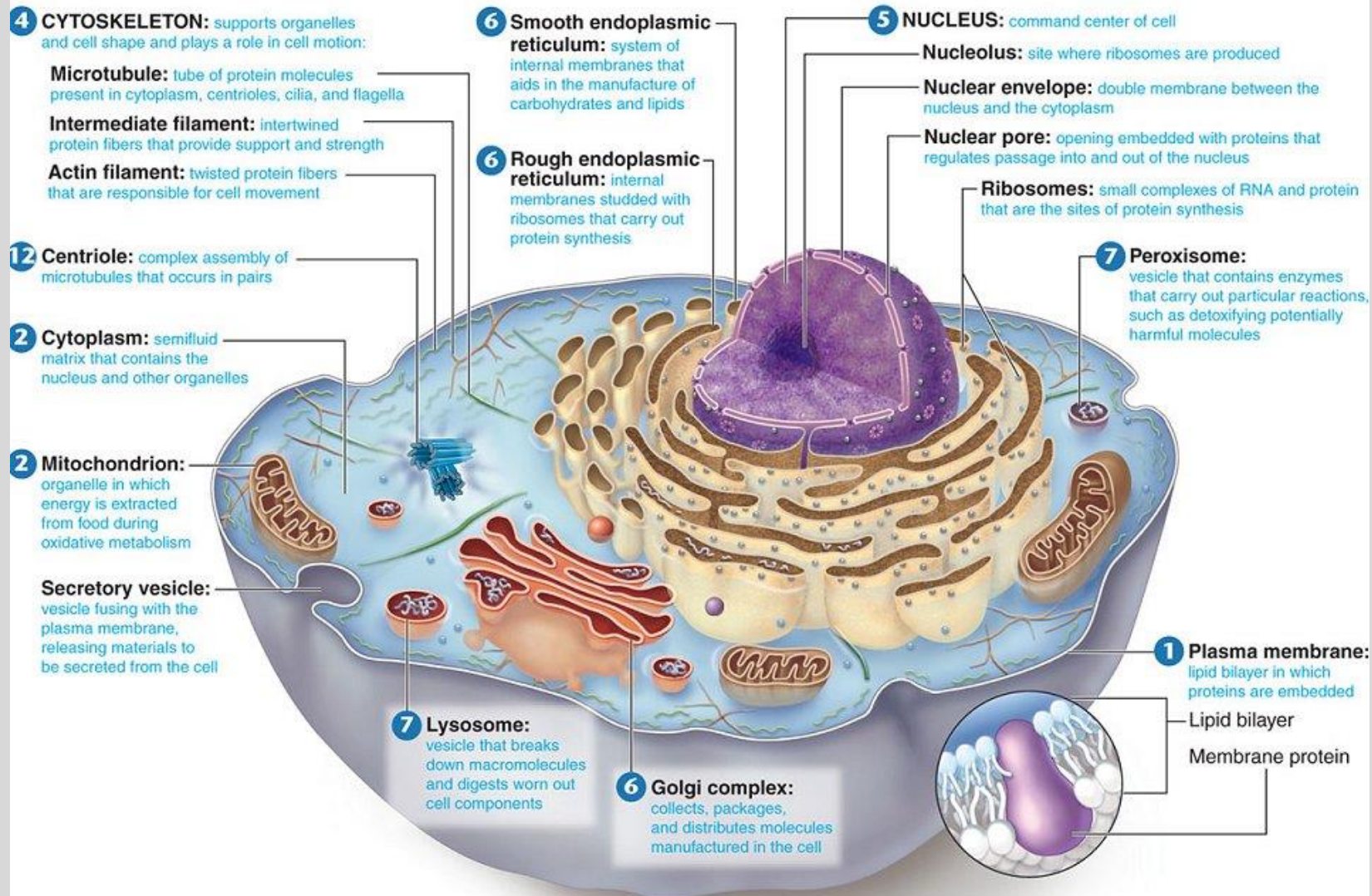
Part 1

By the end of this session students should be able to:

- Recognize cell interactions
- Recognize and enlist causes of cell injury.
- Identify biochemical mechanisms involved in cell injury.

Cell – intro recall

- Basic unit of structure and function
- To maintain their structure and function cells have “homeostasis”
 - Homeostasis is a dynamic condition in which cells try to maintain their internal environment within narrow range of physiologic parameters
- Three main types depending on activity in cell cycle(mitotic ability)
 - Labile- continuous growth without stimulus
 - Stable- growth with stimulus
 - Permanent- no growth even with stimulus
- Every cell type has different ability to withstand the stress and its own homeostatic limits according to tissue function



Normal cell

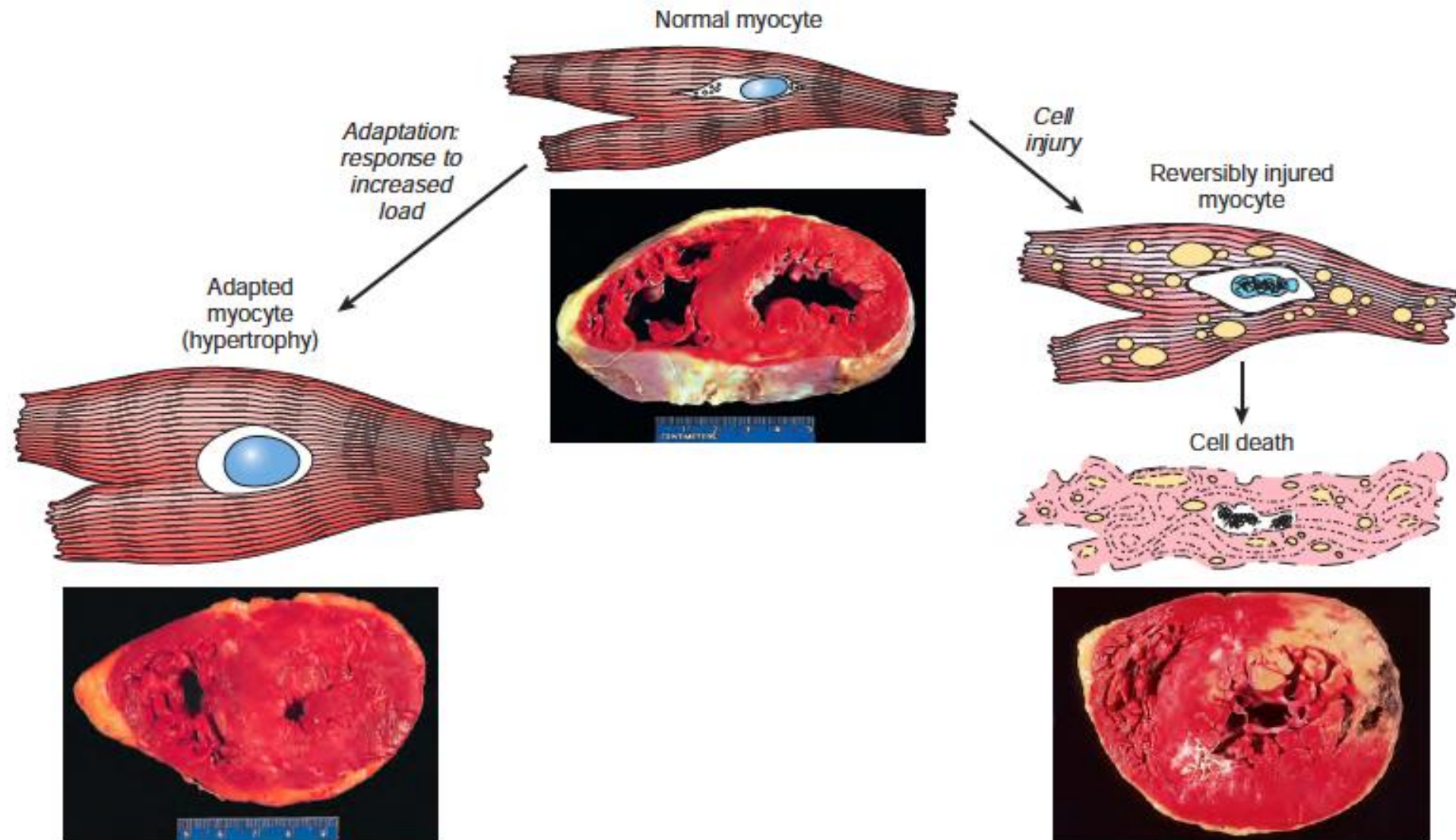
The Cell and the Environment

Adaptive Responses: when exposed to physiological stresses and pathological stimuli, cells undergo adaptations to achieve a new steady state to preserve viability and function

- Atrophy
- Hypertrophy
- Hyperplasia
- Metaplasia

Cell Injury: Occur in response of harmful stimuli when cells are unable or exceed limits to adapt or external stress is severe enough, then cell injury ensues

- Reversible
- Irreversible



Cell stress and injury

- In its specific microenvironment under physical conditions cells survive via homeostasis
- A response from the cellular mechanisms to the stress, such that homeostatic mechanisms were challenged to the limits
- Responses depends upon
 - **Type** of cell
 - **Duration** and **Intensity** of stressor

Cell Injury- affect of the stress on the cell

If confronted with stress that endangers the structural or functional integrity of the cell, the cell may undergo changes to survive depending upon:

- Cell type (Labile/ Stable/ Permanent)
- Duration (removed- mins/hrs/days/years)
- Intensity (mild/ moderate/severe)

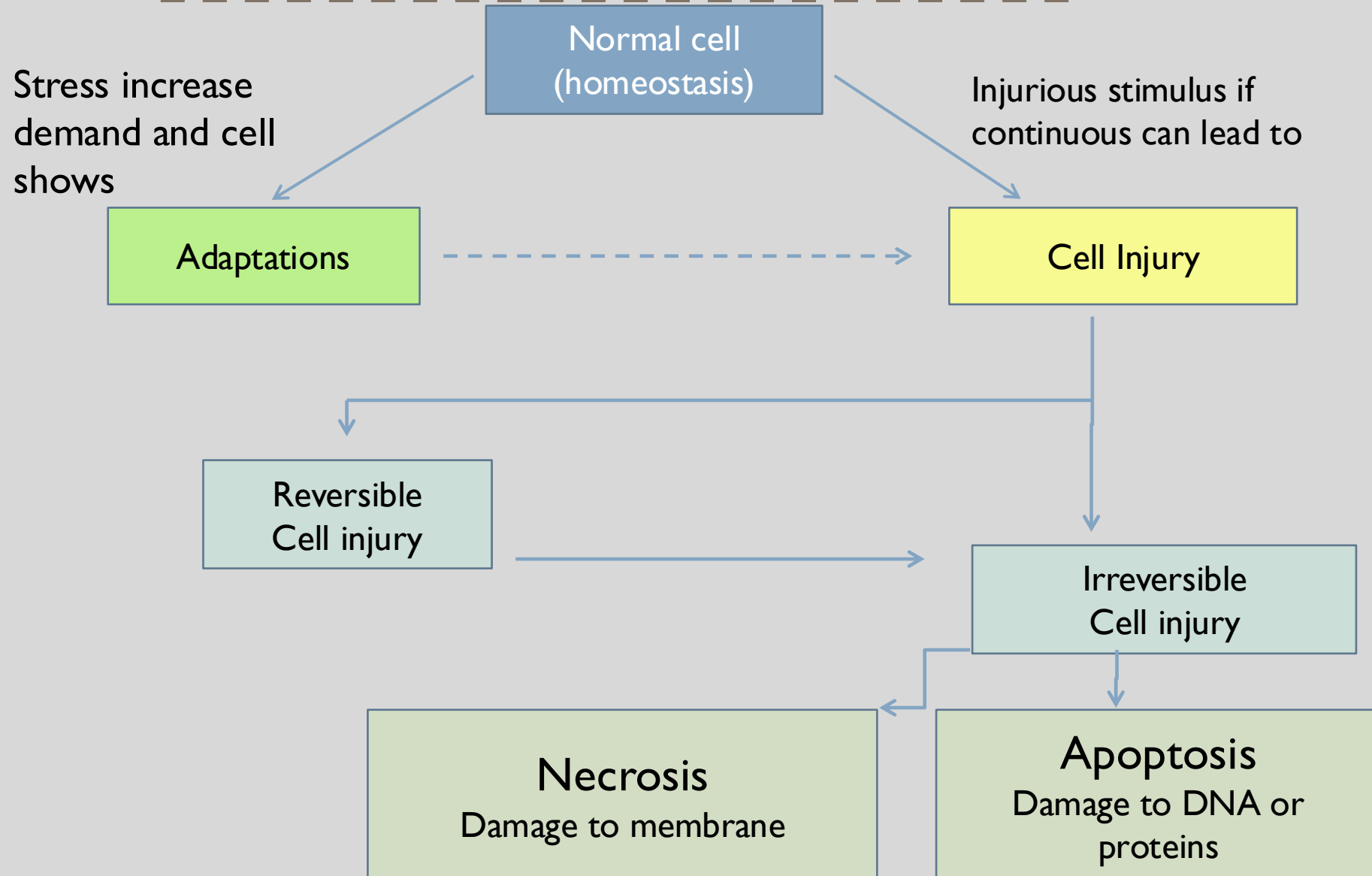
Difference between stress and injury

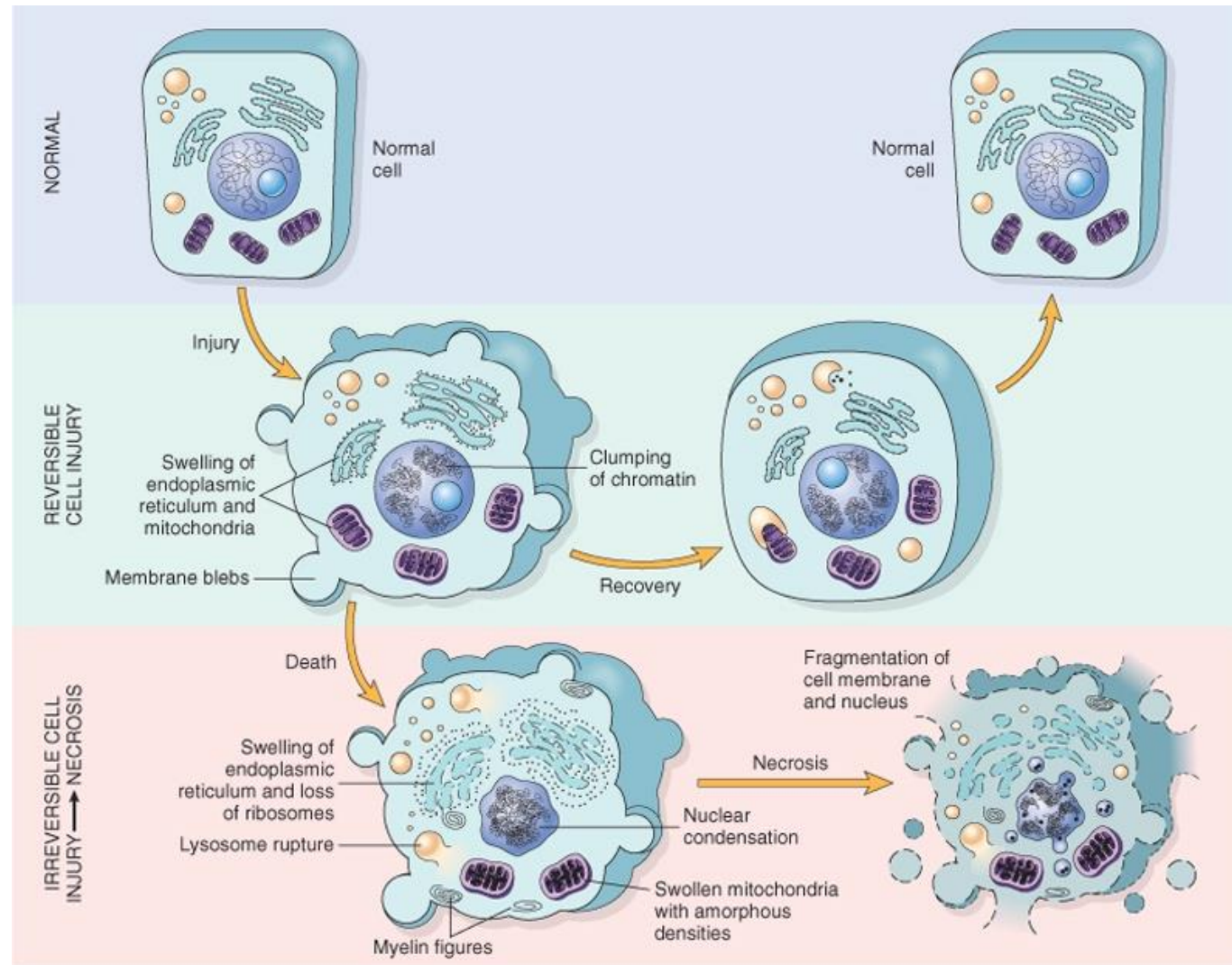
- They are cause (stress) and effect (injury).
 - Removable stressor-removed stress- reversible injury
 - Persistent stressor-sustained stress- Irreversible injury

Cell Injury

- Reversible cell injury:
 - Within certain limits, functional & morphological (gross and microscopic) changes are reversible if
 - damaging stimulus is removed and cell returns to stable state.
 - No membrane damage & nucleus is intact
- Irreversible cell injury (Cell Death):
 - If stress is severe, persistent and rapid then cell unable to restore normal function and progress eventually to cell death.
- Cell Death either:
 - Necrosis
 - Apoptosis

Stages in the cellular response to stress and injurious stimuli





Causes of Cell Injury

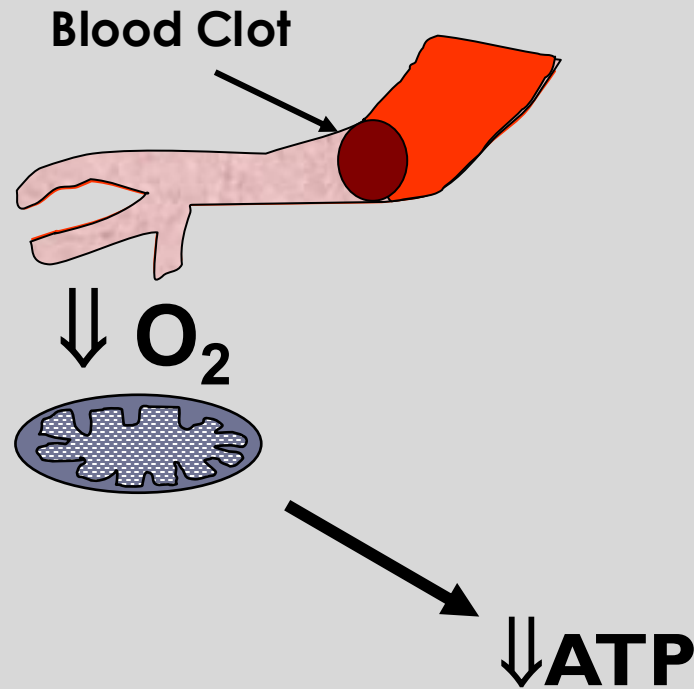
- **Oxygen deprivation-** Hypoxia (most common), ischemia
- **Physical-** sharp (needles, blades, knives, glass etc)/ blunt (steering wheel, falls, pressure etc)
- **Chemical-** pH (alkali/ acid), pollutants
- **Biological-** infections (virus/bacteria/fungi), infestation (parasites or insects/ worms etc)
- **Environmental-** temperature (hypo/hyper), radiation
- **Nutrition-** macronutrient or micronutrient (excess or deficiency)
- **Genetic-** disorders like Down syndrome, Sickle cell anemia, Mediterranean disease
- **Aging-**

General Principles of Cell Injury

- Cellular response to injurious stimuli depends on:
 - Type, duration, and severity of injury.
- Consequences of cell injury depend on:
 - Type, state, & adaptability of injured cell.
- The most important targets of injurious stimuli are:
 - Mitochondria, the sites of ATP generation
 - Cell membranes (homeostasis of cell depends)
 - Endoplasmic reticulum/Ribosomes (Protein synthesis)
 - Cytoskeleton
 - Genetic apparatus of the cell

Mechanism of cell injury

1. Depletion of ATP

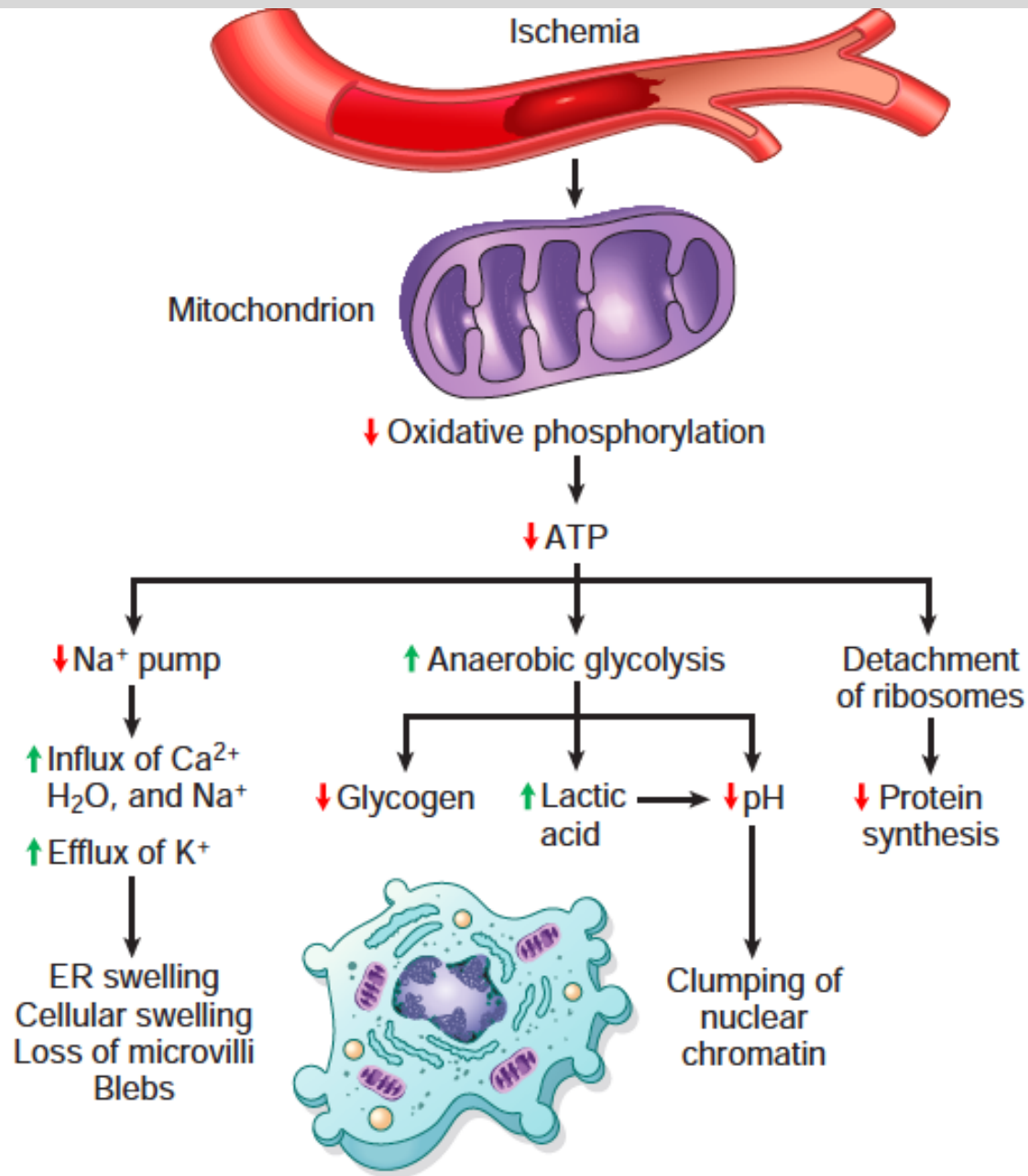


↓ *ATP is the main cause of necrotic cell death*

Major causes of ATP depletion:

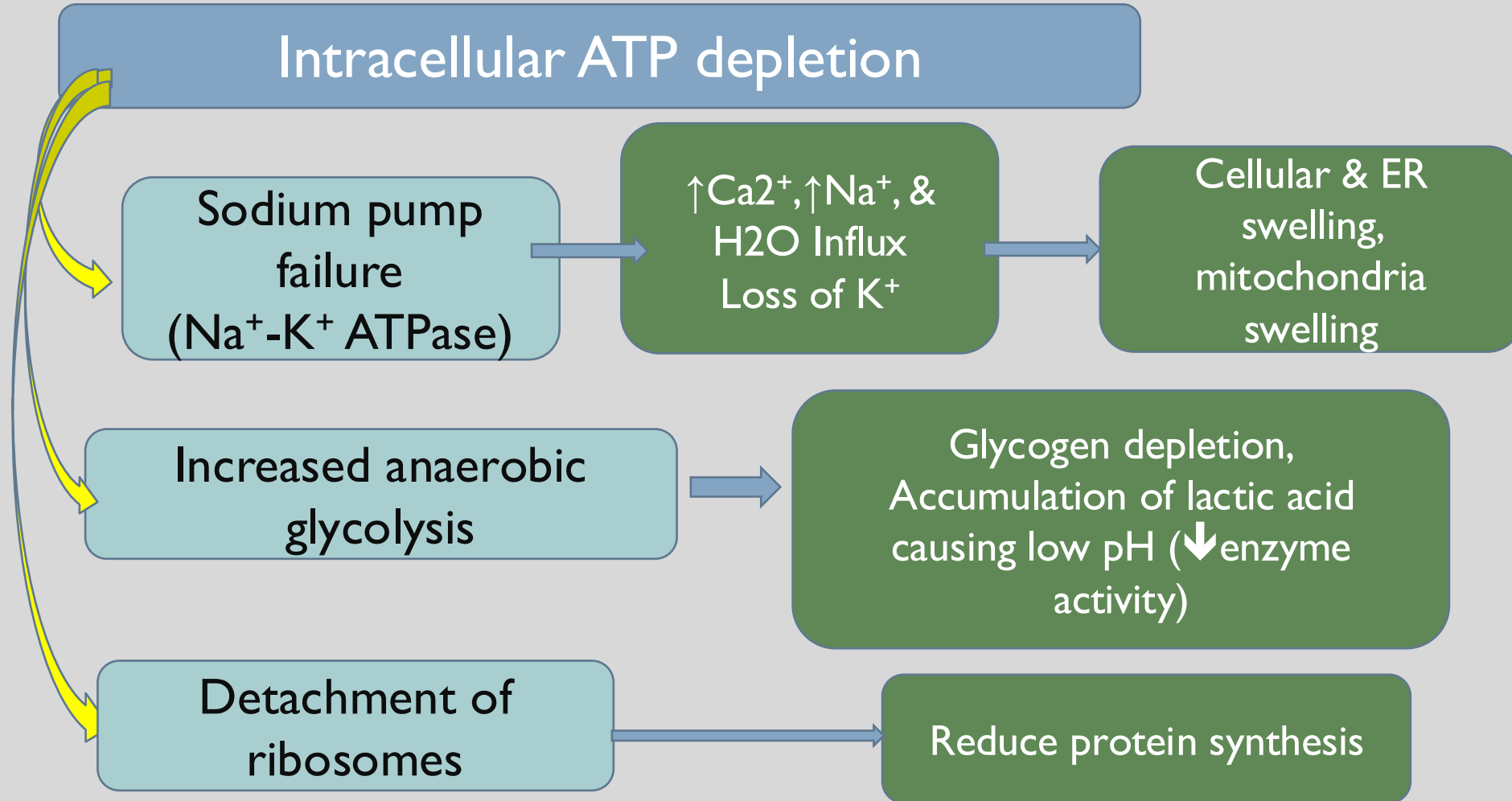
- Reduced supply of oxygen, hypoxia.
- Reduce supply of nutrients
- Mitochondrial damage
- Toxins (Cyanide)

Hypoxic injury is the commonest type of cell injury making hypoxia the commonest factor



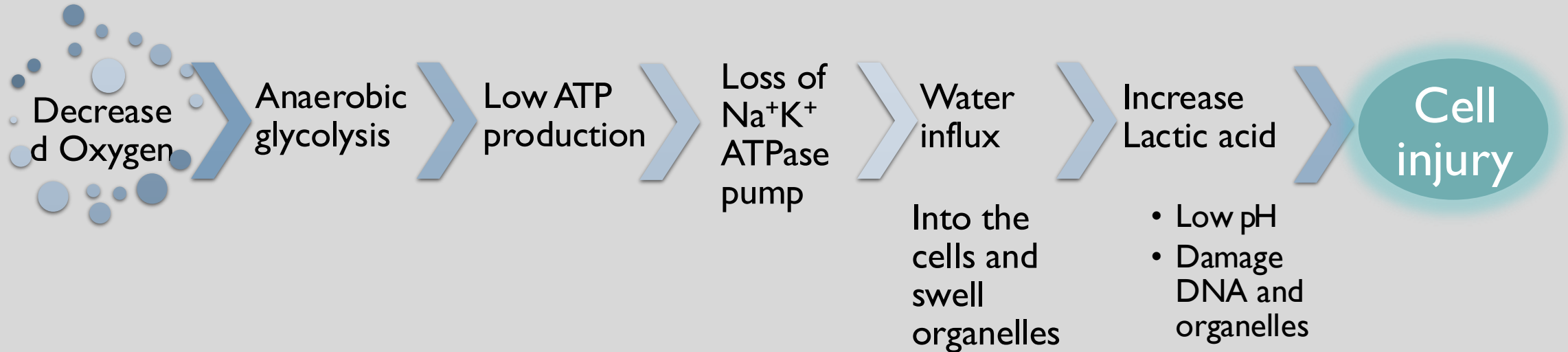
1. Mechanism of Ischemic / Hypoxic injury

Mechanism of Ischemic / Hypoxic injury



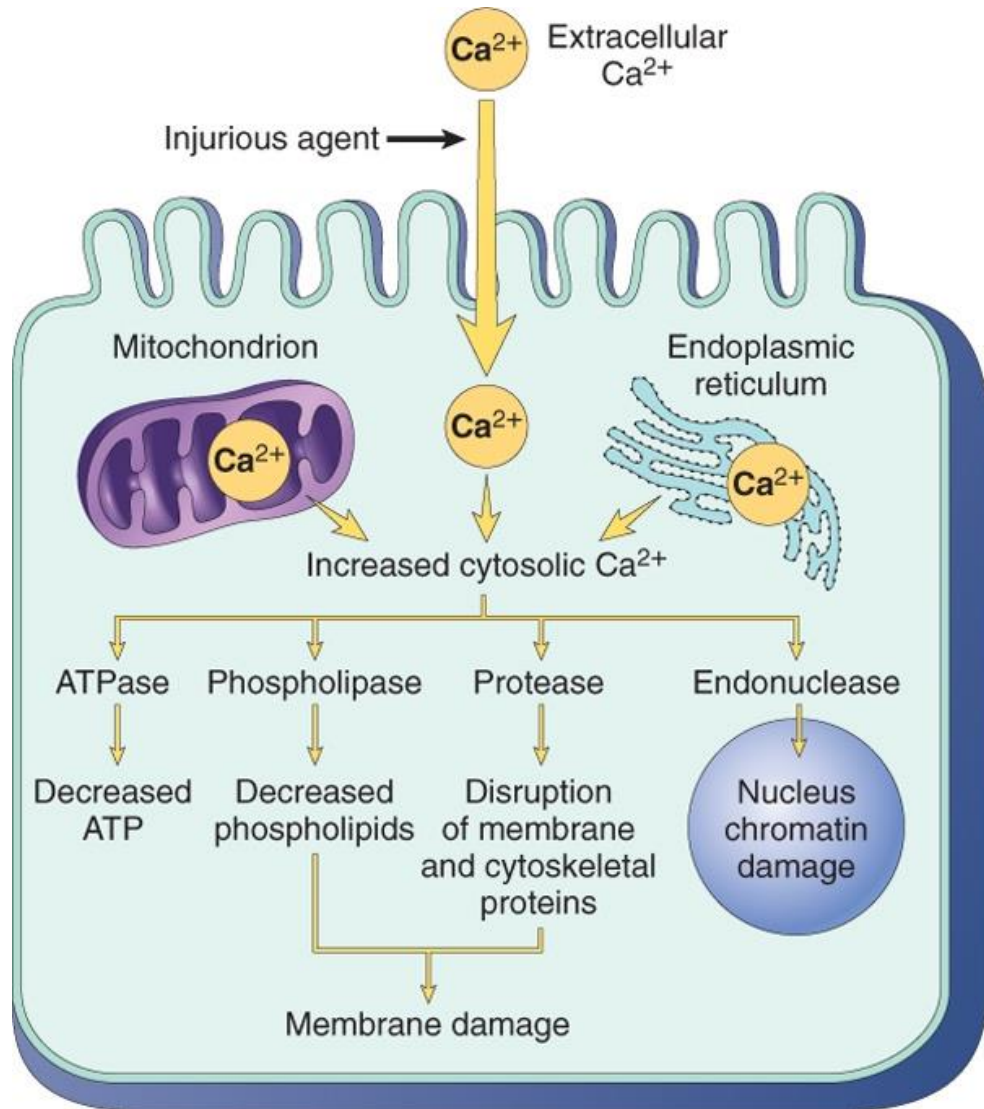
Pathogenesis of hypoxic cell injury

Hypoxia is the most common cause of cell injury



Outline of pathogenesis of hypoxic induced cell injury

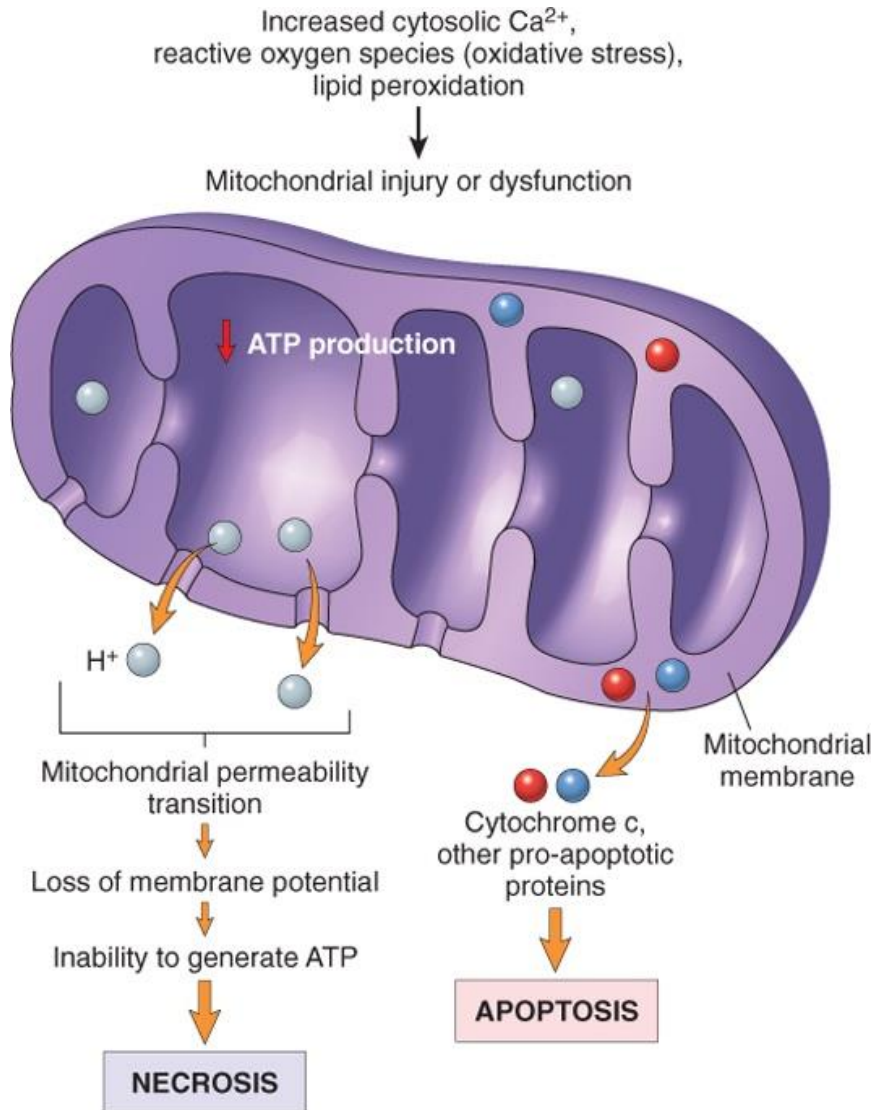
Mechanism of cell injury



2. Influx of Calcium:

- Causes of increase cytosolic Ca^{+2} :
 - Ischemia
 - Toxins
- Influx of Ca^{+2} in cells can:
 - Activate enzymes which can damage the cells (Phospholipase, ATPase, protease, endonuclease)
 - ↓ ATP generation
 - Induce Apoptosis (Caspases)

Mechanism of cell injury



3. Damage to Mitochondria

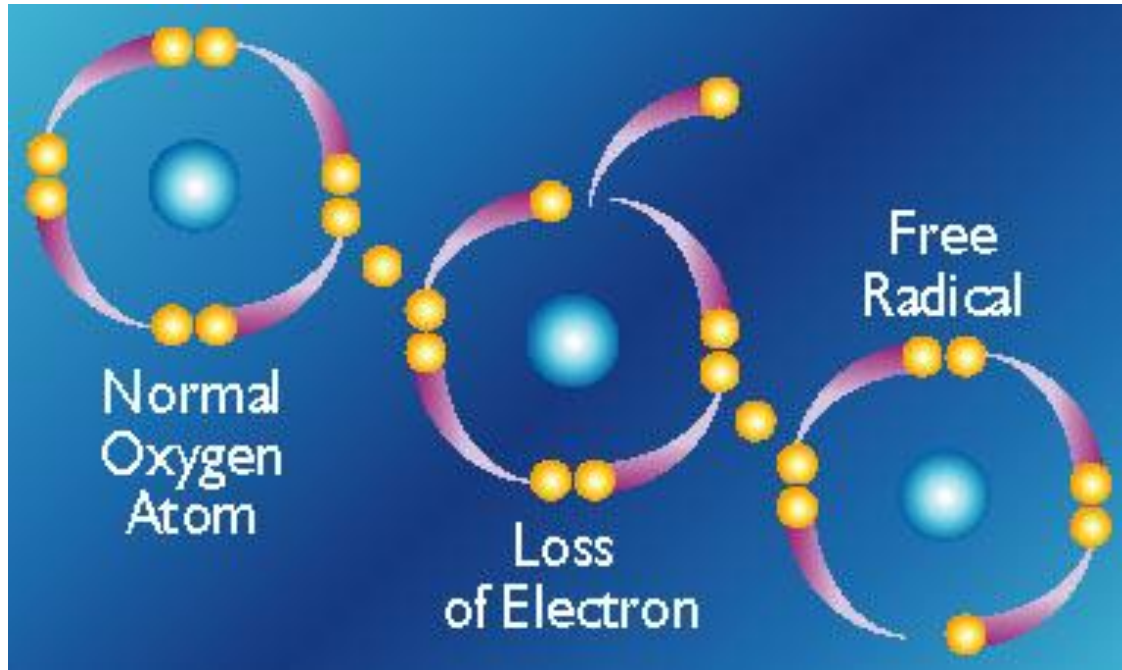
◦ Causes of mitochondrial injury:

- ↑ cytosolic Ca^{+2}
- Reactive oxygen species
- Ischemia

◦ Consequences:

- Formation of channels in mitochondrial membrane
 - ↓ oxidative phosphorylation and depletion of ATP leading to necrosis
 - Release of cytochrome c inducing apoptosis

Mechanism of cell injury

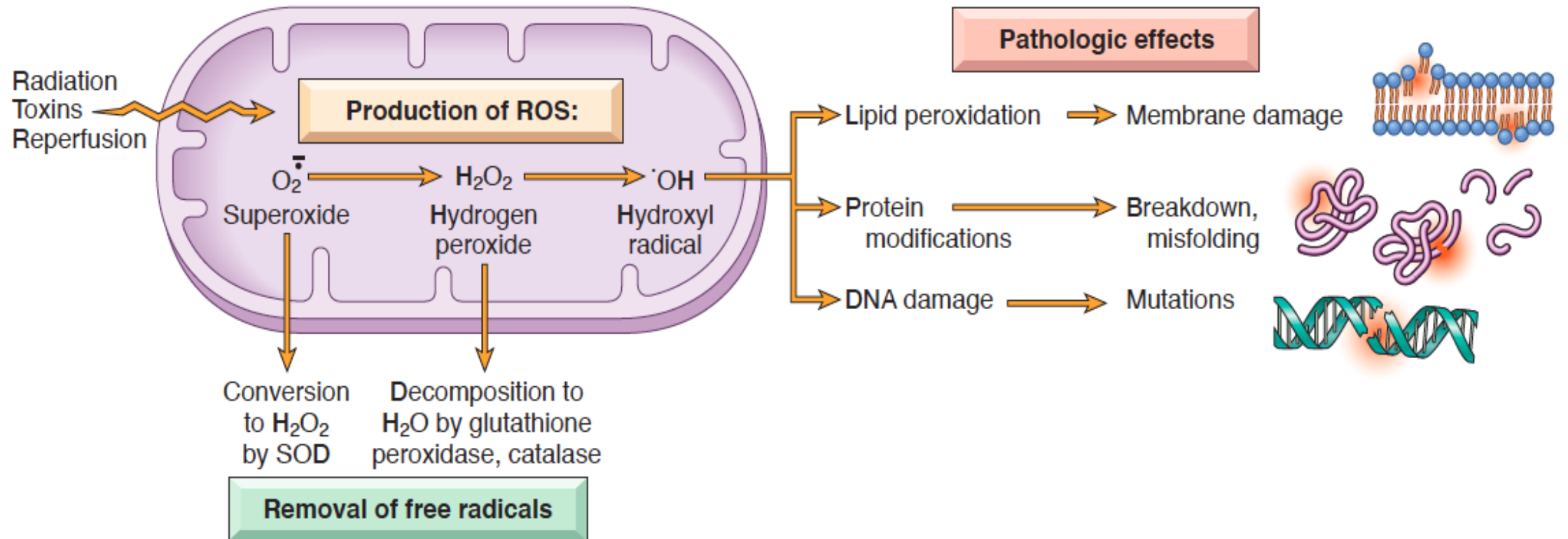


4. Accumulation of Oxygen derived Free Radicals-oxidative stress

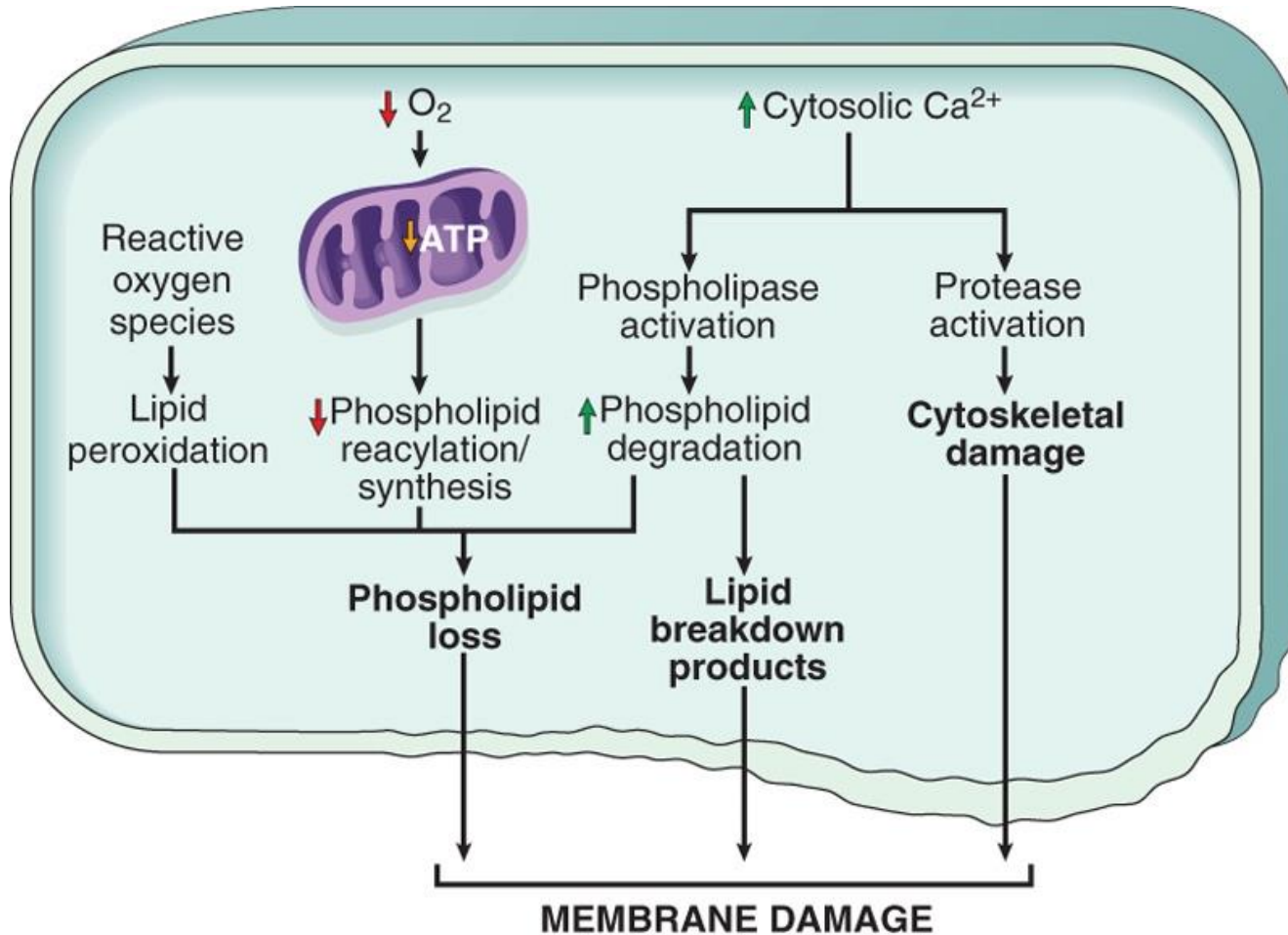
- Free radicals are unstable molecules, which react with cell organelles and cause damage.
- Reactive oxygen species (ROS) are O_2 derived free radicals

Fate of free radicals

- Spontaneous degradation
- Antioxidant action-
Enzymatic degradation
 - SOD (superoxide dismutase)
 - Catalase
- Antioxidant action- Non-enzymatic degradation
 - Carotenes (e.g., Vitamin A)
 - Tocopherols (e.g., Vitamin E)
 - Ascorbate (vitamin C)
 - Glutathione (in liver)
 - Flavonoids (in tea)
 - Selenium
 - Zinc



Mechanism of cell injury



5. Defects in membrane permeability

- Factors that can lead to defect in membrane permeability
 - Hypoxia ($\downarrow$ ATP)
 - ROS (Lipid peroxidation)
 - Increase Ca^{+2} (Activation of enzymes)
- Most important site of membrane damage during cell injury:
 - Mitochondrial membrane damage
 - Plasma membrane damage
 - Injury to lysosomal membrane which release digestive enzymes and degradation of cellular contents

Mechanism of cell injury

6. Damage to DNA and Proteins

- Cells can repair minor DNA damages
- If the damage is severe that cannot be corrected, the cell initiates its suicide program and die by apoptosis.
- Causes of DNA damage:
 - Radiation and chemical injury
 - Reactive oxygen species

Mechanism of cell injury, *summary*

1. ATP depletion (mitochondrial injury) → Na⁺ pump failure → reversible cell injury → necrosis.
2. Mitochondrial injury → necrosis or apoptosis
3. Influx of Ca⁺² activation of enzymes → cell damage/ apoptosis
4. Reactive Oxygen species(free radicals) → damage to cell structure especially DNA.
5. Increase membrane permeability → mitochondria, Lysosome, and cell membranes → necrosis
6. Accumulation of damaged DNA → apoptosis

Summary

THANKYOU

